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REVIEW

Liquid biopsy in triple-negative breast cancer: unlocking the potential of precision oncology

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In the era of precision oncology, the management of triple-negative breast cancer (TNBC) is rapidly changing and becoming more complicated with a variety of chemotherapy, immunotherapy, and targeted treatment options. Currently, TNBC treatment is based on prognostic and predictive factors including immunohistochemical biomarkers [e.g. programmed death-ligand 1 (PD-L1)] and germline *BRCA* mutations. Given the current limitation of existing biomarkers, liquid biopsies may serve as clinically useful tools to determine treatment efficacy and response in both the (neo)adjuvant and metastatic settings, for detecting early relapse, and for monitoring clonal evolution during treatment. In this review, we comprehensively summarize current and future liquid biopsy applications. Specifically, we highlight the role of circulating tumor cell characterization, circulating tumor DNA, and other preclinical liquid biopsy technologies including circulating exosomes, RNA liquid biopsy, and circulating immune-based biomarkers. In the near future, these biomarkers may serve to identify early disease relapse, therapeutic targets, and disease clonality for patients with TNBC in the clinical setting.

Key words: liquid biopsy, triple-negative breast cancer, precision oncology, novel biomarkers, circulating tumor DNA, circulating tumor cells

INTRODUCTION

Current triple-negative breast cancer biomarkers are limited

Triple-negative breast cancer (TNBC) is characterized by the absence of estrogen receptor (ER), progesterone receptor, and human epidermal growth factor receptor 2 (HER2). TNBC represents ∼10%-15% of all BCs, is more common in women younger than 40 years, and is the most aggressive and highly heterogeneous of all BC subtypes.[1]

Despite considerable efforts to identify novel prognostic and predictive biomarkers in TNBC, only a few have been proven useful in clinical trials.[2] These include the germline *BRCA1/2* mutation, present in ∼10%-20% of TNBC cases and leading to homologous recombination deficiency[3-6]; activation of phosphoinositide 3-kinase (PI3K)/protein kinase B (AKT)/mammalian target of rapamycin (mTOR)-dependent pathway; and PD-L1. Among metastatic TNBC (mTNBC) cases, ∼20%-38% of patients express PD-L1.[7,8] The prognostic and predictive roles of elevated tumor-infiltrating lymphocyte (TIL) counts are well established, with their presence being associated with a better prognosis,[9-11] improved response to immunotherapy, and higher rates of pathologic complete response (pCR).[10,12] Importantly, prognosis also depends on lymphocyte localization, with the best outcomes observed in patients with TILs infiltrating both the tumor epithelium and stroma. By contrast, TILs restricted to the tumor margins or absent altogether are associated with the worst prognosis.[13] Besides, the International Immuno-Oncology Biomarker Working Group recently demonstrated the favorable prognostic role of TIL abundance in BC tissue in patients with early-stage TNBC who did not receive adjuvant or neo-adjuvant chemotherapy (NAC).[14]

Recent advancements in noninvasive liquid biopsy have demonstrated its potential role as a predictive biomarker in TNBC. We analyzed current applications of liquid biopsy as prognostic and predictive biomarkers in TNBC across various settings, highlighting it potential as a potential noninvasive, clinically useful tool. In addition, we explore future liquid biopsy applications, including technologies such as circulating exosomes, RNA liquid biopsy, and circulating

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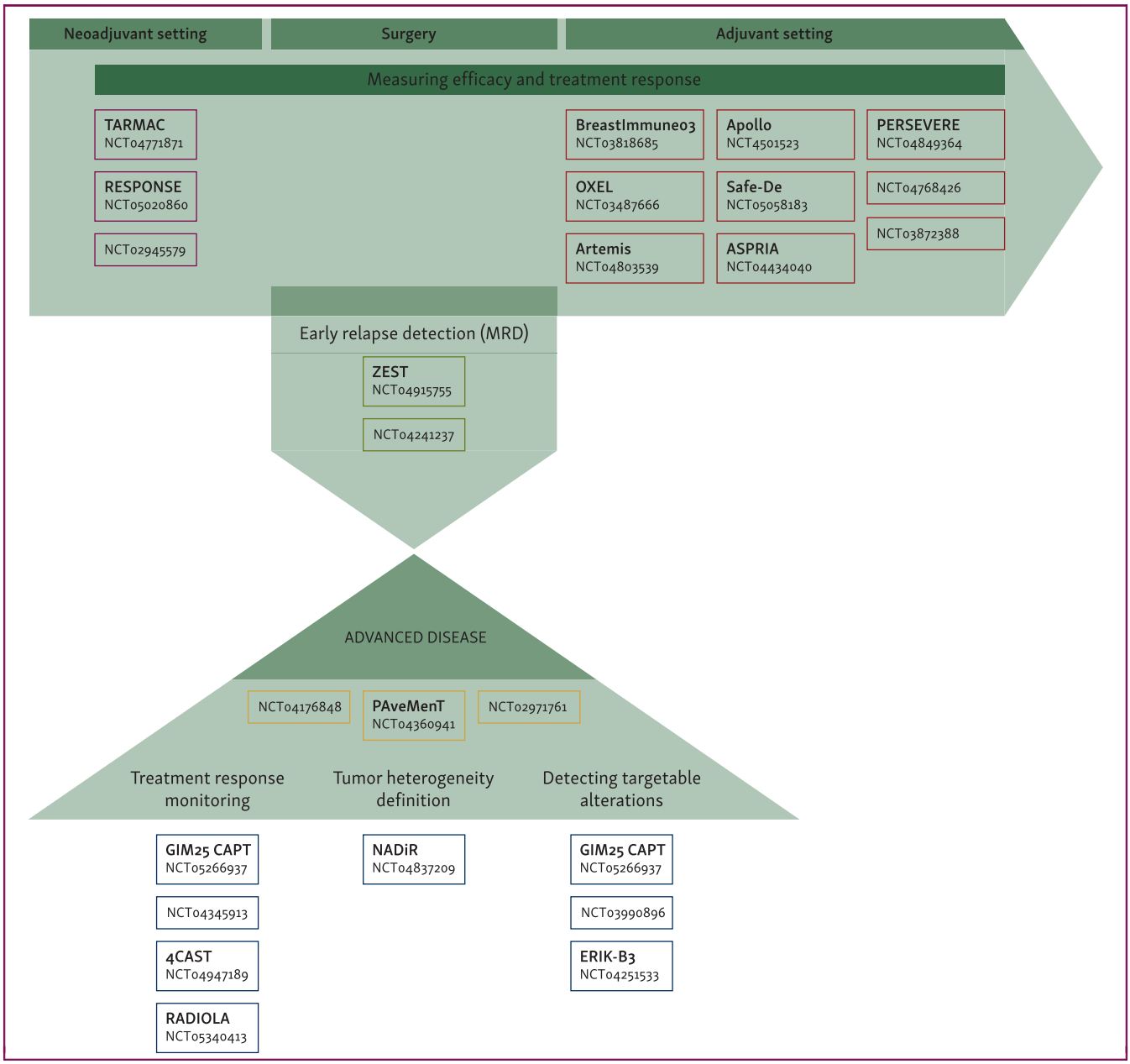

Figure 1. Visual summary of liquid biopsy clinical trials in triple-negative breast cancer.

immune-based biomarkers. A summary of ongoing clinical trials exploring liquid biopsy applications in TNBC are reported in Figure 1 and Tables 1 and 2.

Liquid biopsy

Liquid biopsy involves the analysis of cell-free DNA (cfDNA), circulating tumor cells (CTCs), and other components such as extracellular vesicles and RNA released from tumor cells into blood, urine, cerebrospinal fluid, or bone marrow. There has been an increasing focus on incorporating liquid biopsies into clinical practice for cancer management. Mutations in cfDNA serve as highly specific markers for cancer, and this gave rise to the term 'circulating tumor DNA' (ctDNA). Plasma ctDNA, obtained through blood sampling, provides genetic information from both the primary tumor and metastatic sites.[15] Notably, the half-life of cfDNA in circulation ranges from 16 min to 2.5 h, making ctDNA analysis a *real-time* snapshot of the disease, useful for targeted treatment selection and monitoring.[1,16,17] The number of CTCs can stratify patients with metastatic BC (mBC) into two prognostic groups: stage IV indolent or stage IV aggressive, based on a threshold of $\geq$5 CTCs per 7.5 ml. Patients in the stage IV indolent group exhibit a longer median overall survival (OS).[18]

CTCS IN EARLY TNBC

Trapp's group evaluated the potential prognostic role of CTCs in the routine follow-up of patients with BC. In the adjuvant SUCCESS A trial, the presence of CTCs was assessed before and two years after chemotherapy in 1087

Table 1. Liquid biopsy clinical trials in early TNBC

Trial	Description	Aims in liquid biopsy	Timing
Serial ctDNA Monitoring During Adjuvant Capecitabine in Early TNBC ClinicalTrials.gov Identifier: NCT04768426	A phase II trial of ctDNA monitoring during adjuvant capecitabine in patients with TNBC and residual disease following standard NAC	- *Primary objective:* To characterize the ctDNA profile of TNBC in participants with residual disease after standard NAC receiving standard-of-care adjuvant capecitabine - *Secondary objective:* To correlate ctDNA levels with genomic features and survival	Starting date 3 February 2021 Estimated duration 5 years
Apollo ClinicalTrials.gov Identifier: NCT04501523	A prospective, phase II trial using ctDNA to initiate post-operation boost therapy after NAC in TNBC	To improve the outcome of patients with TNBC using ctDNA to identify those with high relapse risk. ctDNA-positive patients will be randomized to receive boost therapy or standard therapy indicated in NCCN guidelines after NAC	Starting date 3 August 2020 Estimated duration 7 years
PERSEVERE ClinicalTrials.gov Identifier: NCT04849364	A phase II circulating tumor DNA enriched, genomically directed postneoadjuvant trial for patients with residual TNBC	Patients with residual TNBC disease after preoperative therapy will be assigned to one of three arms based on plasma ctDNA positivity and genomic marker(s)	Starting date 24 August 2021 Estimated duration 13 years
ZEST ClinicalTrials.gov Identifier: NCT04915755	A randomized phase III double-blinded study comparing the efficacy and safety of niraparib with placebo in participants with either HER2− BRCA-mutated or TNBC with molecular disease based on the presence of circulating tumor DNA after definitive therapy	Efficacy and safety comparison of niraparib with placebo in participants with HER2− breast cancer susceptibility gene mutation (BRCAmut) or TNBC) with molecular disease (ctDNA) following surgery or completion of adjuvant therapy	Starting date 28 June 2021 Estimated duration 8 years
Safe-De ClinicalTrials.gov Identifier: NCT05058183	Safe de-escalation of chemotherapy for stage 1 BC	- To assess the rates of circulating tumor DNA (ctDNA) in patients treated with surgery for stage 1 BC that is HER2+ or TNBC	Starting date 5 June 2023 Estimated duration 6 years
ASPRIA ClinicalTrials.gov Identifier: NCT04434040	A single-arm phase II trial of atezolizumab with Sacituzumab Govitecan to prevent recurrence in TNBC	- To determine if a combination of two drugs, Sacituzumab Govitecan and atezolizumab, works as a treatment for residual cancer in the breast or lymph nodes and have circulating tumor DNA in the blood	Starting date 2 July 2020 Estimated duration 5 years
TARMAC ClinicalTrials.gov Identifier: NCT04771871	One-stage phase II of treatment response and microRNA profiles in patients with TNBC receiving standard chemotherapy	- To assess the response rate and toxicity of epirubicin−cyclophosphamide with paclitaxel−carboplatin and examine the potential of using circulating microRNA and CTCs as a surrogate marker of chemotherapy resistance in Nigerian women with TNBC	Starting date 29 November 2021 Estimated duration 2 years
Artemis ClinicalTrials.gov Identifier: NCT04803539	A prospective, phase II trial using ctDNA to initiate post-operation boost therapy after adjuvant chemotherapy in TNBC	- To identify patients with TNBC with positive ctDNA and initiate boost therapy in these high-risk patients	Starting date 1 April 2021 Estimated duration 7 years
BreastImmune03 ClinicalTrials.gov Identifier: NCT03818685	A multicenter, randomized, open-label phase II study to evaluate the clinical benefit of a post-operative treatment associating radiotherapy + nivolumab + ipilimumab versus radiotherapy + capecitabine for patients with TNBC with residual disease after NAC	*Secondary outcome measures:* - ctDNA detection and analysis - Molecular subtyping (RNA-seq)	Starting date 2 July 2019 Estimated duration 5 years
OXEL ClinicalTrials.gov Identifier: NCT03487666	A pilot study of immune checkpoint or capecitabine or combination therapy as adjuvant therapy for TNBC with residual disease following NAC	Secondary outcome measures: - Quantification of ctDNA at different timepoints	Starting date 21 July 2018 Estimated duration 4 years
RESPONSE ClinicalTrials.gov Identifier: NCT05020860	A phase II trial to correlate early clinical response to pathologic outcome with neoadjuvant systemic therapy in patients with EBC	- To determine whether a decrease in ctDNA levels (as measured from baseline to surgery) correlates with clinical and/or pathologic response	Starting date 18 April 2023 Estimated duration 6 years
ClinicalTrials.gov Identifier: NCT03872388	Atorvastatin in treating patients with stage IIb-III TNBC who did not achieve a pathologic complete response after receiving neoadjuvant chemotherapy	- *Primary objectives:* To determine the proportion of patients with undetectable CTCs at 6 months among those with stage IIB/III TNBC who did not achieve a pCR or residual cancer burden-I after receiving NAC with and without atorvastatin therapy	Starting date 14 January 2019 Estimated duration 4 years
ClinicalTrials.gov Identifier: NCT02945579	Eliminating surgery or radiotherapy after systemic therapy in treating patients with HER2+ or TNBC	*Other outcome measures:* - Change in biomarkers in blood and plasma	Starting date 20 January 2017 Estimated duration 9 years

CTC, circulating tumor cell; ctDNA, circulating tumor DNA; NAC, neoadjuvant chemotherapy; NCCN, National Comprehensive Cancer Network; pCR, pathologic complete response; TNBC, triple-negative breast cancer.

Table 2. Liquid biopsy clinical trials in metastatic TNBC

Trial	Description	Aims in liquid biopsy	Timing
GIM25 CAPT ClinicalTrials.gov Identifier: NCT05266937	A phase II trial of atezolizumab plus carboplatin plus paclitaxel as first-line therapy in patients with metastatic PD-L1-positive TNBC	- Variation of ctDNA levels from the baseline to the first evaluation through the multigene FoundationOne Liquid NGS panel - Variation of ctDNA levels from the first evaluation to progression through the multigene FoundationOne Liquid NGS panel - Association between archival PD-L1 levels and target genes expression levels through CLIO analysis - Variation of target gene expression levels through the CLIO analysis from the baseline to the first evaluation - Variation of target gene expression levels through the CLIO analysis from the first evaluation to disease progression	Starting date 3 July 2020 Estimated duration 4 years
ClinicalTrials.gov Identifier: NCT03990896	A phase II clinical trial to evaluate the effectiveness of talazoparib as a potential treatment for metastatic breast cancer with a *BRCA1* or *BRCA2* mutation	- Studying the efficacy of talazoparib in 30 patients with mBC who have pathogenic somatic *BRCA1/2* mutations detected in cfDNA	Starting date 18 November 2021 Estimated duration 3 year
EPIK-B3 ClinicalTrials.gov Identifier: NCT04251533	A phase III, multicenter, randomized, double-blind, placebo-controlled study to assess the efficacy and safety of alpelisib (BYL719) in combination with nab-paclitaxel in patients with advanced TNBC with either *PIK3CA* mutation or *PTEN* loss without *PIK3CA* mutation	- To determine whether treatment with alpelisib in combination with nab-paclitaxel is safe and effective in patients with advanced TNBC who carry either a *PIK3CA* mutation (study part A) or have *PTEN* loss (study part B1) or PTEN loss without PIK3CA mutation (study part B2). *PIK3CA* mutation determined by ctDNA	Starting date 8 June 2020 Estimated duration 7 years
ClinicalTrials.gov Identifier: NCT04345913	A phase I/II trial evaluating the safety and efficacy of eribulin in combination with copanlisib in patients with mBC	- To determine ctDNA mutation profiles at baseline and changes in mutation profile and VAFs on cycle 2 day 1 (C2D1) and at disease progression compared with baseline to correlate with treatment response - To assess circulating biomarkers predictive of treatment response - To assess plasma and serum proteomics and metabolomics predictive of treatment response	Starting date 1 March 2021 Estimated duration 3 years
NADiR ClinicalTrials.gov Identifier: NCT04837209	A phase II study of niraparib, dostarlimab, and radiotherapy in metastatic, PD-L1 negative or immunotherapy-refractory TNBC	Other outcome measures: - To evaluate changes in ctDNA using a patient-specific NGS ctDNA assay	Starting date 21 July 2021 Estimated duration 8 years
4CAST ClinicalTrials.gov Identifier: NCT04947189	A phase Ib dose exploration and dose expansion, open-label, single-center study evaluating the safety and efficacy of INO-464 in combination with chemotherapy in patients with mBC	- Gene expression analysis (RNA-seq, or single-cell RNA-seq) on tumor biopsies in response to study treatments - Expression of androgen receptor, ZEB1, and other protein markers in tumor biopsies in response to study treatment - Gene expression and/or ctDNA analysis in blood and/or tumor biopsies in response to the study treatments	Starting date 1 November 2021 Estimated duration 4 years
PAveMenT ClinicalTrials.gov Identifier: NCT04360941	A phase Ib study of palbociclib and avelumab in metastatic AR+ TNBC	- Exploratory assessment of ctDNA suppression as a potential biomarker of response - Investigate changes in peripheral blood mononuclear cells in patients receiving avelumab and palbociclib and the relationship with treatment response - Exploratory assessment of biomarkers of response in baseline tumor biopsy, and of acquired resistance in progression biopsies and plasma DNA	Starting date 11 August 2020-08-11 Estimated duration 4 years
ClinicalTrials.gov Identifier: NCT02971761	A phase 2 clinical trial of the combination of pembrolizumab and selective androgen receptor modulator GTX-024 in patients with metastatic androgen receptor-positive TNBC	- To evaluate the effect of the combination therapy on peripheral blood CTCs and ctDNA - To evaluate the effect of combination therapy on TEX and TEX-associated immune biomarkers	Starting date 1 June 2017 Estimated duration 5 years

Continued

Table 2. Continued

Trial	Description	Aims in liquid biopsy	Timing
ClinicalTrials.gov Identifier: NCT04176848	A phase II study of CFI-400945 and durvalumab in patients with advanced/metastatic Triple Negative Breast Cancer (TNBC)	*Primary outcome measures:* Objective response rate of CFI-400945 given with durvalumab using RECIST 1.1 (timeframe: 24 months) *Secondary outcome measures:* - Disease control rate of CFI-400945 given with durvalumab (timeframe: 24 months) - Immune-related response rate (iRECIST) of CFI-400945 given with durvalumab (timeframe: 24 months] - Safety and tolerability of CFI-400945 given orally in combination with durvalumab assessed by CTCAE (timeframe: 24 months) - Immune effects of CFI-400945 + durvalumab measured in cfDNA (timeframe: 24 months)	Starting date 19 December 2019 Estimated duration 3 years
RADIOLA ClinicalTrials.gov Identifier: NCT05340413	Predicting olaparib sensitivity in patients with unresectable locally advanced/metastatic HER2-negative breast cancer with BRCA1, BRCA2, PALB2, RAD51C or RAD51D mutations or RAD51-foci low test	*Secondary outcome measures:* Capacity of the ctDNA drop after 4 weeks of treatment to predict the efficacy of olaparib in both cohorts	Starting date 25 March 2022 Estimated duration 2 years

AR, androgen receptor; cfDNA, cell-free DNA; CTCAE, Common Terminology Criteria for Adverse Events; ctDNA, circulating tumor DNA; mBC, metastatic breast cancer; NGS, next-generation sequencing; PD-L1, programmed death-ligand 1; TEX, tumor-derived exosome; TNBC, triple-negative breast cancer; VAF, variant allele frequency.

patients with high-risk BC.[19] Two years post-chemotherapy, 198 patients were CTC positive, and this found to be an independent, statistically significant prognostic factor for poor OS and disease-free survival (DFS). CTC status at the 2-year follow-up was independent of baseline CTC status. Moreover, patients who were CTC positive both at baseline and the 2-year follow-up had the worst OS and DFS.[20]

Measuring treatment response in the neoadjuvant and adjuvant setting

Several studies have investigated the clinical validity of CTCs in assessing the efficacy of NAC and treatment response. In patients with nonmetastatic TNBC, the presence of one or more CTCs after completing NAC was associated with significantly decreased OS and relapse-free survival[21] Bidard et al.[22] presented the results of a meta-analysis based on 21 studies that detected CTCs by CELLSEARCH in 2030 patients with early BC (EBC) before NAC and surgery, 25.8% of whom had TNBC. The authors demonstrated the prognostic role of CTCs in patients with EBC treated with NAC, independent of tumor subtypes, thereby enhancing current prognostic models.

CTC detection after surgery or during adjuvant therapy has been associated with a poor prognosis. To investigate the role of CTC monitoring in early TNBC, 286 women were enrolled in the study[23] Considering CTC levels after surgery, patients with >5 CTCs per 7.5 ml of blood experienced worse outcomes compared to patients with <5 CTCs. Specifically, patients with higher CTC counts had a recurrence rate of 22.4% seven days post-surgery.

CTCs IN METASTATIC TNBC

The prognostic value of CTC enumeration has been demonstrated in metastatic BC (mBC) patients using several studies with the Food and Drug Administration (FDA)-cleared CELLSEARCH (Menarini Silicon Biosystems) methodology. However, multiple technologies are available for the detection and enumeration of CTCs.[18,24]

The molecular characteristics of CTCs contribute to the formation of metastatic lesions, enabling these cells to invade tissues, survive in circulation, and extravasate to remote locations.[24] In TNBC, multicellular CTC clusters are associated with a worse outcome compared to single CTCs. Specifically, CTC clusters can promote the development of metastatic disease 20 to100 times more frequently than a single CTCs.[25]

The presence of CTCs in patients with TNBC varies by stage, with a higher number of CTCs observed in mBC patients. Patients with CTCs at or above the threshold of 5 per 7.5 ml of blood before treatment, and those who fail to clear the cells during treatment, have a significantly worse outcome compared to those who maintain a CTC count of less than 5 after starting systemic therapy.[26] Another study showed that patients with CTCs ≥5 per 7.5 ml, and/or those who experienced an increase in CTC counts 3-5 weeks and/or 6-8 weeks after the start of treatment had decreased PFS and OS.[27] However, the SWOG S0500 trial did not demonstrate improved clinical outcomes in any mBC subtypes, including TNBC, when using early CTC-guided treatment changes.[28] In a retrospective analysis, investigators classified patients into three prognostic sub-groups based on to baseline CTC enumeration, confirming that patients with higher CTC counts per 7.5 ml of blood had worse outcomes.[29]

In the TBCRC 001 trial, a direct comparison of two CTC enumeration methods, CELLSEARCH and IE/FC, showed higher concordance between the techniques. CTC enumeration 7-14 days after treatment initiation was correlated with time-to-progression, suggesting that CTCs may serve as an early marker of response to targeted therapy and a more reliable indicator of progression risk compared to baseline counts.[30]

Detecting targetable alterations

Analysis of CTCs also provides insights into the molecular characteristics of patients with mTNBC. Abreu et al.[31] immunoisolated CTCs from a cohort of 32 patients with stage III and IV TNBC were analyzed using CELLSEARCH technology. The samples were characterized with a panel of genes related to cancer aggressiveness and plasticity. The expression signature identified in CTCs was associated with a hybrid epithelial−mesenchymal transition (EMT) status and a stem-like phenotype. These cells were detected in 42% of patients, and 3 of these patients also had CTC clusters. All CTC-positive patients were metastatic at the time of sample collection, and 26% of these patients had >5 CTCs. No correlation was found between the number of CTCs and other clinicopathologic features. Patients with >5 CTCs had worse PFS and OS. Cellular plasticity in CTCs from patients with TNBC, indicated by the expression of hybrid EMT and stem cell markers, was associated with poor prognosis and increased aggressiveness of these tumors.[31]

PIK3CA is an actionable cancer gene in mBC, as previously demonstrated in hormone receptor positive (HR+) mBC.[32] Pestrin et al.[33] analyzed the *PIK3CA* mutational status within single CTCs isolated from 39 patients with mBC, 20 of whom had samples enriched with ≥5 CTCs. *PIK3CA* mutations were identified in six patients, and discordance between the *PIK3CA* status of the primary BC (wild type) and the matched CTC (exon 20 mutation) was observed in only one patient. This suggests a proof of concept for the potential utility of liquid biopsy in this setting.

Detecting tumor heterogeneity

The characteristics of CTCs change during tumor cell spreading, mainly due to the EMT process, which together with expression of stemness markers could facilitate chemotherapy resistance and promote their capacity to metastasize.[31] Rothé et al.[34] investigated mBC heterogeneity by analyzing CTCs matched with synchronous tumor biopsies from three patients with mBC. Considering tumor mutational burden (TMB) in tumor biopsies in the patient with TNBC, 38% of all single-nucleotide variants (SNVs) found in the CTC samples were also present in tumor biopsies. Interestingly, all SNVs in cancer driver genes identified in tumor biopsies were also detected in CTCs. Single CTC genomic analysis revealed some driver aberrations not found in the tumor sample and provided additional insights into the clonal/subclonal distribution of all aberrations identified.

ctDNA IN EARLY TNBC

Prognostic value

In locally advanced TNBC, there has been a continual escalation of treatment approaches, as supported by two landmark trials, the KEYNOTE-522 trial[35] and the CREATE-X trial in the adjuvant setting.[36] In this context, an emergent need is treatment tailoring in early TNBC according to the risk of recurrence,[37] as already expressed in the St. Gallen International Expert Consensus Conference in 2017.[38] ctDNA and CTCs were considered potentially useful biomarkers to guide treatment de-escalation.[37]

Notably, ctDNA may be detectable in early BC using high-sensitivity minimal residual disease (MRD) assays. The ctDNA detection after surgical resection reflects the persistence of micrometastatic residual disease, which is otherwise clinically undetectable.[39,40] Therefore ctDNA could potentially serve as a marker of residual disease in patients with TNBC to guide therapeutic decisions after neoadjuvant therapy. Based on this potential, the Q-CROC-03 trial examined patients with TNBC undergoing NAC with biopsies carried out before chemotherapy and after chemotherapy as well as blood samples before, during, and after NAC to determine molecular factors of response or resistance to standard NAC. A slight increase in ctDNA levels was predictive of incomplete pCR, and the absence of ctDNA levels at this presurgical specimen was associated with long-term relapse-free survival and OS, with a prognostic value similar to pCR status.[41]

Measuring efficacy of neoadjuvant therapy

Further information can be obtained from liquid biopsy to anticipate NAC response. Magbanua et al.[42] provided evidence regarding the role of ctDNA as a predictive biomarker for response and outcome in the I-SPY 2 trial, collecting blood samples before treatment (T0), 3 weeks after paclitaxel initiation (T1), between paclitaxel and anthracycline regimens (T2), and before surgery (T3). At T0, 73% of the patients had detectable ctDNA, which decreased over time (T1 35%; T2 14%; and T3 9%). Patients who remained ctDNA positive at T1 were significantly more likely to have residual disease after NAC (83% non-pCR) compared with those who cleared ctDNA (52% non-pCR). All patients who achieved pCR after NAC were ctDNA negative. In the non-pCR group, ctDNA-positive patients had a significantly increased risk of metastatic recurrence; notably, ctDNA-negative patients had excellent outcome, similar to those who achieved pCR. This suggests that rising ctDNA before surgery may be an indicator of increased risk of relapse and may characterize patients who might benefit from treatment intensification before surgery to achieve a pCR and prevent distant relapse.

Early relapse detection

ctDNA detection after surgery, along with CTC enumeration, may be used to detect MRD and assess which patients may later develop disease recurrence. In a cohort of 55 women with EBC at risk of cancer relapse treated with NAC independently of BC subtypes, plasma samples were collected at baseline, 2-4 weeks after the operation, and then every 6 months during follow-up. MRD was detectable before standard clinical relapse using mutation tracking in serial samples, which increased sensitivity for predicting clinical relapse with a median lead time of 7.9 months.[43] Importantly, the genetic events of the primary tumor may differ from those found in the metastatic disease.[44] To date, many

companies are investing in the development of liquid biopsy platforms to detect ctDNA in the MRD setting.[45] In this context, two different approaches exist: the tumor-informed approach looking for targeted variation [Signatera, Natera (platform)] and the tumor-uninformed approach searching *de novo* variations [Guardant Reveal, Guardant Health (assay)].[46] The tumor-informed approach requires *a priori* knowledge of the tumor profile by tumor tissue analysis, and the MRD assay is patient specific, but this approach could potentially miss clonal variants different from the primary tumor sample. In the tumor-uninformed approach, a broad next-generation sequencing panel is used to look for MRD combined with methylation analysis, but background noise from nontumor-derived mutations could influence the results.[45]

Tumor-informed approaches with personalized PCR were utilized in the following trials to detect MRD starting from tumor tissue analysis. In a cohort of 170 patients with EBC, detection of MRD was associated with future clinical relapse for all major BC subtypes, with ctDNA detected before relapse in 22 of 23 patients.[47] In a prospective trial of 42 patients with TNBC, ctDNA detection after NAC was associated with increased relapse risk, regardless of other clinicopathological features. Considering recurrent cases, ctDNA was detected 13 months before clinical diagnosis.[43] In another trial, samples from 38 patients with early-stage TNBC with matched tumor and plasma were analyzed. Of the 33 patients who had a mutation identified in their primary tumor, mutations in ctDNA were detected in 4 patients, who experienced a rapid recurrence. Next-generation sequencing of ctDNA in patients with TNBC with residual disease after NAC could predict recurrence with high specificity but moderate sensitivity.[48] With the cTRAK TN trial, Turner et al.[49] confirmed that early ctDNA detection was associated with high rates of metastatic disease. The trial was unable to address the co-primary endpoint of pembrolizumab's effect on ctDNA clearance due to only few patients receiving the programmed cell death protein 1 (PD-1) inhibitor treatment. However, the trial demonstrated the clinical critical need to have more frequent timepoints and higher sensitivity assays in ctDNA detection. Given the timing of plasma sampling and assay sensitivity, the cTRAK TN trial did not validate the approach that ctDNA MRD could be detected with sufficient lead time before radiographic evidence of disease to impact clinical management. This demonstrates the urgent need to optimize sampling timepoints and assay performance, and also, in parallel, develop more effective therapies for micrometastatic disease in the future. This is perhaps a minor setback in the field, but the assay utilized was far from optimal.

ctDNA IN METASTATIC TNBC

Prognostic value

In a retrospective trial of 164 patients with mTNBC, the presence of a ctDNA fraction >10% was associated with worse outcomes, regardless of clinicopathological data.[50] The strong prognostic tumor-agnostic impact of ctDNA was demonstrated in terms of calculating the circulating tumor fraction.[51] It was also demonstrated that similar patterns of chromosomal alterations exist in primary and mTNBC, suggesting that most somatic copy number alterations occur early in tumorigenesis in TNBC.[52]

Detecting targetable alterations

Plasma ctDNA genotyping can identify mutations in potentially 'actionable' cancer genes.[53] In a cohort of patients with mBC, considering all subtypes, with identified *BRCA1/2* mutations using cfDNA genotyping, 13.5% of patients had *BRCA1/2* somatic mutations, while 4% had known germline mutations. Sensitivity to poly (ADP-ribose) polymerase inhibitors (PARPis) in CTC-derived models was also explored. Some cell lines with somatically acquired driver variants showed increased sensitivity to PARPi, but others did not, suggesting differential response to PARPi based on *BRCA1/2* mutations detected using cfDNA.[54] This is further being evaluated in a phase II clinical trial to assess talazoparib efficacy in 30 patients with mBC presenting with pathogenic somatic *BRCA1/2* mutations detected in cfDNA.[55]

In the plasmaMATCH trial,[56] ctDNA was used to direct therapy in patients with mBC to screen for rare actionable mutations. Among the 1051 patients, 17% of patients had an mTNBC, with a small subset of patients receiving neratinib for a *HER2* mutation or capivasertib for an AKT pathway activating mutation. In cohort E, including patients with TNBC without targetable mutation and treated with olaparib plus ceralasertib (ataxia telangiectasia and Rad3-related kinase - inhibitor), response rate to treatment did not meet prespecified criteria for efficacy.[57] However, this trial demonstrated sufficient promise for ctDNA testing to guide selection of targeted therapies.[56]

Detecting tumor heterogeneity

Liquid biopsy could also detect intrapatient variability with respect to genomic alterations of primary and metastatic lesions. In a large clinically annotated cohort of patients with mBC who underwent ctDNA evaluation, the genomic landscape of patients with mTNBC (22% of population) was described, demonstrating genetic heterogeneity of mBC in blood.[58] The mutant allele frequency (MAF) of the highest variant in the TNBC cohort was significantly higher compared with HR+ or HER2 positive (HER2+) patients. The most common SNVs were *TP53* and *PIK3CA*, and the most frequent copy number variants were *MYC*, *CCNE1*, and *PIK3CA*. Confirming these data, Page et al.[59] reported SNVs detected in *ESR1* in three patients with mTNBC to provide evidence for small ER+ clones as a potential resistance mechanism. These data support the concept of increasing tumor heterogeneity in mTNBC compared with other mBC subtypes, with liquid biopsy serving as a potentially useful clinical tool. Data from a retrospective cohort of patients with mBC highlight the role of ctDNA in monitoring tumor genetic evolution. Increasing MAF (≥3.3) and number of alterations (≥4) were associated with disease progression.[16]

COMPLEMENTARY ROLE OF CTDNA AND CTC ENUMERATION

Early relapse detection in early TNBC

Detection of ctDNA and CTCs in patients with early-stage TNBC after NAC was independently associated with disease recurrence and could represent a stratification factor for future postneoadjuvant trials. In the BRE12-158 trial that randomized 196 patients with early-stage TNBC with residual disease after NAC to receive postneoadjuvant genomically directed therapy versus treatment of physician's choice, blood samples were collected at the time of treatment assignment. Overall, the presence of ctDNA and CTC after NAC was associated with significantly inferior distant DFS and OS.[60]

However, there is still an unmet need to establish a CTC threshold for recurrence, improve MRD detection sensitivity, and determine the optimal timepoints for blood sample collection.[61] In 2022, the ESMO Precision Medicine Working Group could not recommend the use of molecular residual disease/molecular relapse detection in clinical practice due to the lack of validated assays for both ctDNA and CTC detection.[62]

Treatment response monitoring and tumor heterogeneity in mTNBC

Recently, the clinical value of concomitant analysis of CTCs and ctDNA was investigated in a large prospective biomarker study (COMET) conducted in 189 patients with HER2 negative (HER2−) mBC treated with first-line chemotherapy, including 23% of patients with mTNBC. To understand the complementary contribution of CTCs and ctDNA, both biomarkers were analyzed before treatment and after 4 weeks of first-line chemotherapy. The CTC enumeration at 4 weeks (≥5 CTC/7.5 ml) after treatment initiation determined the best prognostic model for PFS with ctDNA variant allele frequency (VAF) at baseline, grade, and tumor subtype. CTCs and ctDNA provided complementary clinical information.[63] However, their role as monitoring tools needs further prospective validation, as underlined in the ESMO recommendations.[62] In addition, Venesio et al.[53] demonstrated the complementary role of ctDNA and CTCs in describing cancer heterogeneity due to substantial differences between primary tumor and metastatic/recurrent disease.

IMMUNOTHERAPY AND LIQUID BIOPSY

The need to develop tumor-based predictive biomarkers is still unmet for patients with TNBC treated with immunotherapy.[2] T-cell expansion and ctDNA analysis are promising tools, while the prognostic and predictive value of CTC enumeration is being explored with respect to PD-L1 expression on CTCs.[64] TMB represents a pan-cancer feature associated with benefits from immunotherapy.[2,64] It is defined by the number of somatic mutations per DNA megabase pair (mut/Mb) and may serve as a surrogate marker for the production of neoantigens, which can lead to increased T-cell infiltration.[2,65] In BC, the role of TMB in tumor immunogenicity remains unclear, as it is generally low to intermediate, and the proportion of patients with high TMB (≥10 mut/Mb, TMB-H) is different across BC groups.[65-67] Tumor whole exome sequencing or gene panel sequencing data from 3969 patients with primary or mBC were analyzed to evaluate the frequency, mutational patterns, and genomic profile of hypermutated BC. Hypermutation was identified in 5% of all cases of BC with enrichment in metastatic tumors (mBC 8.4% versus primary BC 2.9%). The median TMB was significantly higher in TNBC (1.8 mut/Mb) compared with HR+ (1.1 mut/Mb) or HER2+ BC (1.3 mut/Mb). According to preliminary data from this study, hypermutated BC would be more likely to benefit from PD-1 inhibitors.[66] However, TMB-H has not been consistently demonstrated as a predictive factor for immune checkpoint inhibitors (ICIs) across all cancer types. This is due to variability in how different tumors are classified as TMB-H according to cancer type,[68] despite the FDA's tissue-agnostic approval for this threshold.[69] Only a small proportion of TNBC cases (8%-10%) have a high TMB. Bianchini et al.[2] suggested improving the predictive value of TMB in TNBC by considering specific mutational signatures, the number of clonal mutations, dinucleotide variants, indels, or additional genomic alterations.

Liquid-based TMB is still under investigation; several studies have explored the role of ctDNA in determining blood-based TMB as a potential clinically actionable biomarker.[70] Considering both pretreatment and on-treatment VAF in blood samples across 16 advanced-stage tumor types from three phase I/II trials of durvalumab (with or without the anti-CTLA-4 therapy tremelimumab), the 'molecular response' was defined using ctDNA to predict long-term survival and identified early responders among patients with initially radiologically stable disease.[71] In the INSPIRE trial, a correlation was demonstrated between baseline ctDNA concentration and clinical response, clinical and survival benefit, with a stronger association utilizing ctDNA dynamics during immunotherapy treatment.[72] In the NIMBUS trial patients with HER2− mBC with tissue TMB-H experienced benefit from nivolumab plus ipilimumab therapy and investigators identified a subgroup of patients with TMB ≥14 Mut/Mb with higher ORR, underlining the unmet need of the optimal TMB cut-off in predicting benefit to ICI treatment in mBC.[73]

In the SAFIR02 BREAST IMMUNO study, *CD274*, the gene encoding PD-L1, emerged as a new biomarker related to PD-L1 expression. This cancer cell-extrinsic feature is associated with greater benefit from ICIs in patients with mTNBC, presenting new challenges and opportunities for liquid biopsy applications.[74]

EMERGING OR PRECLINICAL LIQUID BIOPSY BIOMARKERS

RNA liquid biopsy targets

MicroRNA (miRNA) is a unique miRNA signature secreted into the bloodstream from the tumor's primary site that could be used as a liquid biopsy target. The benefits of using

miRNA-based assays include insight into specific genetic information throughout tumorigenesis in multiple bodily fluids due to their stability throughout the circulation.[75] These targets have been investigated in several preclinical trials of patients with BC but have not been extensively evaluated in clinical trials. One of the earliest studies attempted to find a unique miRNA signature in the serum of patients with BC independent of subtype, and reported changes in miRNA expression in patients diagnosed with BC compared with healthy controls.[76] These data supported the preclinical findings of varying RNA expression in multiple types of BC cell lines *in vitro*, including an increase in specific miRNA in TNBC cell lines.[77] Assessment of cell-bound miRNA showed unique miRNA signatures in peripheral blood mononuclear cells (PBMCs) due to their supportive effect in the tumor microenvironment.[78] There were different miRNA expressions of PBMCs between triple-positive and TNBC cells. The miRNA in the PMBCs were different from the miRNA profile from the tumor site. Cell-free miRNA signatures were different from each other in luminal BC and TNBC, creating the possibility of differentiating tumors by liquid biopsy.[79] Alterations in miRNA from tumor and tumor supporting cells are also observed, but there is no consensus on which miRNA changes are specific to define a new TNBC diagnosis.

Another potential application of miRNA liquid biopsies is to serve as a prognostic biomarker after diagnosis. For example, an increase in total serum RNA in nonmetastatic BC was found compared with mBC. In addition, metastatic tumors were more likely to express a different miRNA signature compared with nonmetastatic cancer[80] and changes in miRNA expression could be used to predict OS in patients with TNBC.[81] The miRNA profile secreted into the bloodstream is likely different, limiting the utility of these signatures in a clinical-based liquid biopsy without additional validation. Besides, miRNA expression was associated with a decrease in OS and an increase in tumor reoccurrence.[82] Multiple miRNA signatures have been identified as potential prognostic biomarkers but without a defined miRNA profile that predicts TNBC progression, limiting their current clinical utility.[83]

Circulating exosomes and liquid biopsy

Another potential biomarker is circulating exosomes, which are exocytosed vesicles secreted by tumor cells into the bloodstream. One major advantage of developing an exosome-based assay for cancer diagnosis and monitoring is the enhanced exosome stability in a variety of environments, such as blood, urine, or saliva.[84] These vesicles package a variety of genetic material and proteins, potentially allowing for analysis of multiple targets from a single sample that mimics the tumor at its origin. Exosomes isolated from patients with BC independent of subtypes have a higher concentration of CD45+ leukocyte-derived microparticles, carcinoembryonic antigen, and CA15-3 compared with healthy people,[85] as well as an increase in the expression of antiapoptotic proteins and their splice variants; additionally, the circulating expression of these proteins correlated with the expression at the primary tumor site.[86] Exosome-bound miRNAs have an advantage over free miRNAs due to further enhanced stability, increased quantity, and more literature supporting their use as a biomarker.[87] Eichelser et al.[88] found an increase in specific miRNA signatures and surface proteins on exosomes in sera collected from patients with HR− BC. Considering 18 different miRNA compounds between TNBC and HER2+ malignancies, 9 had a differential expression between patients with BC and healthy people. Elevated miRNA levels in sera collected from TNBC and HER2+ BC correlated with patients who were more likely to have pCR.[89] Patients with TNBC had upregulation of specific exosomal miRNA compared with healthy controls. Besides, there was a difference in the miRNA expression of patients who had a BC recurrence after treatment and those who did not develop recurrence.[90] Exosomal miRNA may be helpful in predicting metastasis: exosome miRNA expression changes between patients with mBC and patients without metastasis.[91] Exosomes can also contain tumor-associated DNA; however, the benefit of measuring tumor DNA in exosomes remains unclear. ctDNA was more sensitive to detecting the presence of DNA expression compared with exosomal DNA.[92] Exosome-based liquid biopsies are currently being evaluated due to their potential role in TNBC diagnosis prognostication and therapeutic monitoring.[93]

Circulating immune-based biomarkers as liquid biopsy targets

A common feature of the tumor microenvironment is the presence of immunomodulators and tumor-associated immune cells. Thus monitoring these changes in the immune response may allow clinicians to monitor the host response against BC.[94,95] Liquid biopsies may enable the investigation of PD-L1 expression heterogeneity in TNBC between the primary site and distant metastatic sites.[96,97] Increased PD-L1 expression on the surface of CTCs was useful in predicting reduced OS in patients with mBC.[98] Other immunogenic proteins have shown differences in patients diagnosed with cancer as well. Increased levels of early myeloid-derived suppressive cells (eMDSC) as well pro-eMDSC proteins in patients with TNBC were associated with a worse response to NAC.[99] Previous studies using all subtypes of BC samples have shown an increase in CD117- and CD11b-expressing granulocytes at the time of diagnosis with BC.[100] Analysis of the serum samples after surgical resection and radiotherapy showed decreased expression, suggesting that reducing the tumor burden reduces the measurable immune response. In addition, a decreased serum concentration of myeloid-derived suppressive cells, PD-L1-expressing T lymphocytes, and regulatory T cells was associated with a clinical benefit across all BC subtypes.[101] Tumor-associated macrophages (TAMs) are recognized as having a significant role in tumorigenesis and immune evasion in BC.[102] Concentrations of circulating CD163-positive TAMs have potential as a future serum immune

cell biomarker.[103] Monitoring serum TAM concentrations could demonstrate immune response to ongoing cancer therapies and help guide treatment with antimacrophage antibody therapies in patients with TNBC.

CONCLUSIONS

Liquid biopsy has the potential to be a useful tool in clinical practice to assess treatment response in all settings, including detecting early relapse and monitoring disease biology in the metastatic setting. We presented the current state of the art and future applications of liquid biopsy in TNBC. There are still significant unmet clinical needs to validate the role of ctDNA and CTCs in identifying therapeutic targets and clonality in TNBC serum markers in the metastatic setting. In addition, further research is needed to explore the role of CTC characterization and other liquid biopsy technologies, such as RNA liquid biopsy, circulating exosomes, and circulating immune-based biomarkers.

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